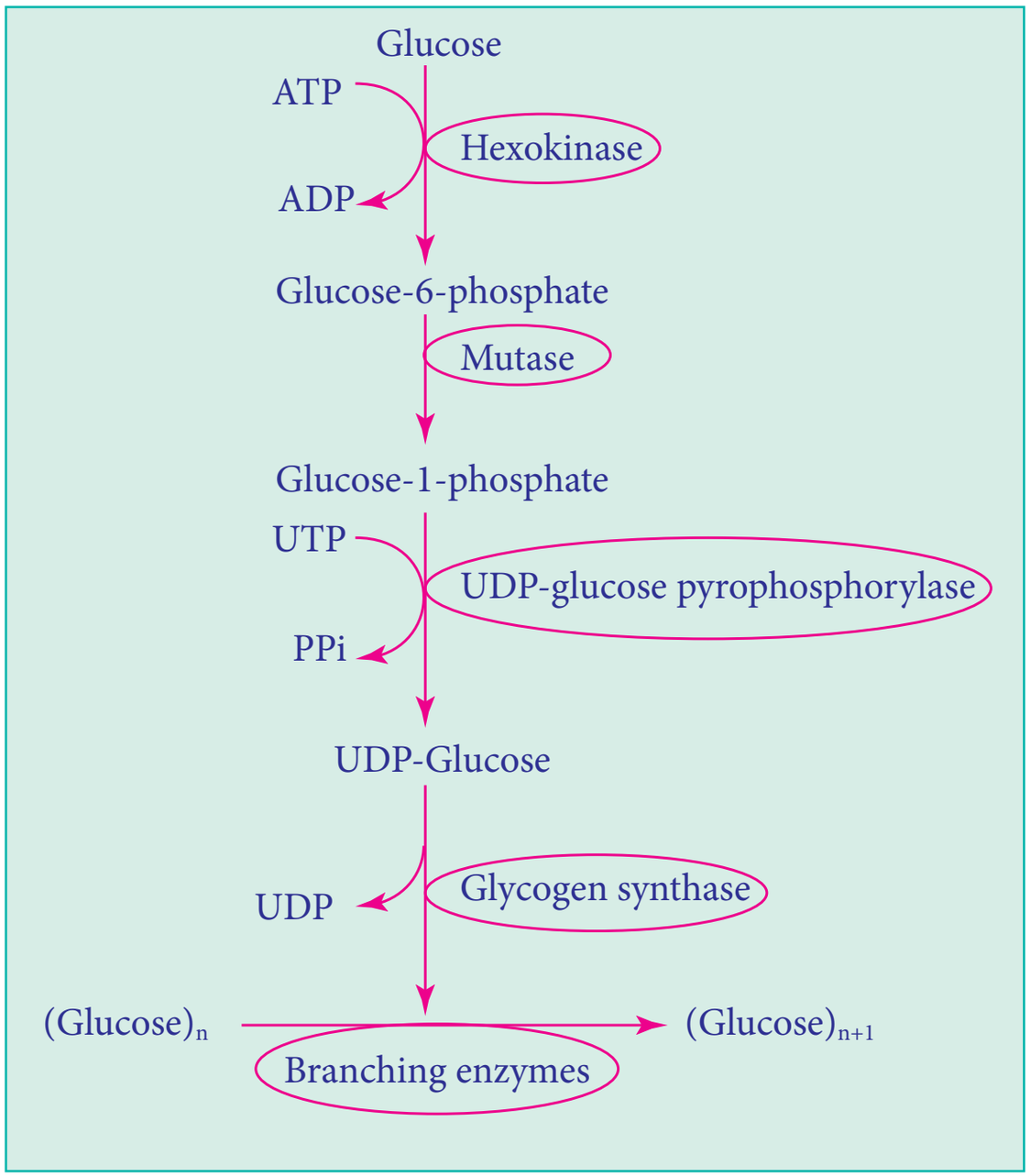


3.6.2. Glycogenolysis

Glycogenolysis is the biochemical breakdown of glycogen to glucose, to provide immediate energy. It takes place in the cells of muscle and liver tissues in response to hormonal and neural signals. In particular, glycogenolysis plays an important role in the adrenaline-induced fight-or-flight response and the regulation of glucose levels in the blood. It is stimulated by the hormones glucagon and epinephrine (adrenaline).

The process of glycogenolysis involves the sequential removal of glucose monomers by phosphorolysis, a reaction catalysed by the phosphorylated 'a' form (active form) of the enzyme glycogen phosphorylase. This enzyme cleaves the glycosidic bond linking a terminal glucose to a glycogen branch by substituting a phosphoryl group for the α [1→4] linkage producing glucose-1-phosphate and glycogen that contains one less glucose molecule.

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